

Data Control Language (DCL) Commands

Q. What is Data Control Language (DCL)? Explain the concept of user privileges and demonstrate the use of GRANT and REVOKE commands.

📌 **Definition to Remember**

Data Control Language (DCL) is a subset of SQL used by **Database Administrators (DBAs)** to control user access and maintain database security. It contains two commands: **GRANT** (give permissions) and **REVOKE** (remove permissions). DCL enforces the **Principle of Least Privilege**.

1. User Privileges and Access Control

A **Privilege** is a specific permission to perform an action on a database object.

Privilege Type	Description	Examples
System Privileges	Administrative tasks	<code>CREATE TABLE</code> , <code>CREATE USER</code>
Object Privileges	Actions on specific tables/views	<code>SELECT</code> , <code>INSERT</code> , <code>UPDATE</code> , <code>DELETE</code>

2. GRANT Command

Gives specific privileges to a user or role.

Syntax:

```
GRANT privilege_name ON object_name TO user_name [WITH GRANT OPTION];
```

- `WITH GRANT OPTION` allows the receiving user to further grant the privilege to others.

Examples:

```
-- Grant SELECT and INSERT on Employee table to user 'john'
GRANT SELECT, INSERT ON Employee TO john;

-- Grant all privileges to a manager role
GRANT ALL ON Employee TO manager_role;
```

3. REVOKE Command

Removes previously granted privileges from a user or role.

Syntax:

```
REVOKE privilege_name ON object_name FROM user_name;
```

Examples:

```
-- Remove INSERT from 'john' (he can still SELECT)
REVOKE INSERT ON Employee FROM john;

-- Remove all privileges from a former employee
REVOKE ALL ON Employee FROM ex_employee_user;
```

★ Must-Write Points (for 10 marks)

1. DCL controls **who can access** which database objects and **what actions** they can perform.
2. Two DCL commands: **GRANT** (give privileges) and **REVOKE** (remove privileges).
3. Two types of privileges: **System Privileges** (admin tasks) and **Object Privileges** (table actions).
4. **GRANT privilege ON object TO user** — grants specific access.
5. **WITH GRANT OPTION** allows the user to further pass the privilege to others.
6. **REVOKE privilege ON object FROM user** — removes a specific privilege.
7. DCL enforces the **Principle of Least Privilege** — give only the access needed.

⚡ Quick Recall

DCL → Security Control → Privileges (System + Object) → GRANT (give access, WITH GRANT OPTION) → REVOKE (remove access) → Principle of Least Privilege